Álgebra en FEC (I)

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$$
\begin{array}{r|l}
 & \quad a^5x + a^{10} \\
\hline
a^3x^2 + a^7x + a^2 & a^8x^3 + a^9x^2 + a^{12}x + a^{12} \\
 & \underline{a^8x^3 + a^{12}x^2 + a^7x} \\
 & a^{13}x^2 + a^2x + a^{12} \\
 & \underline{a^{13}x^2 + a^2x + a^{12}} \\
 & 0
\end{array}
$$

Resumen

Ejercicio 1 Cambio de representación y suma

a) $\boxed{\alpha^3 + \alpha^9 = \alpha}$

b) $\boxed{\alpha^{11} + \alpha^{13} = \alpha^4}$

c) $\boxed{\alpha^3 + \alpha^9 + \alpha^{11} + \alpha^{13} = 1}$

d) $\boxed{\alpha^5 + \alpha^5 = 0}$

Ejercicio 2 Producto y división

a) $\boxed{\alpha^6 \cdot \alpha^{11} = \alpha^2}$

b) $\boxed{\alpha^{12} \cdot \alpha^9 = \alpha^6}$

c) $\boxed{\dfrac{\alpha^3}{\alpha^{10}} = \alpha^8}$

d) $\boxed{\dfrac{1}{\alpha^7} = \alpha^8}$

e) $\boxed{(\alpha^{14})^3 = \alpha^{12}}$

Ejercicio 3 Expresiones mixtas

a) $\boxed{\dfrac{(\alpha^5 + \alpha^9) \cdot \alpha^7}{\alpha^2 + \alpha^{12}} = \alpha^6}$

b) $\boxed{(\alpha^4 + 1)^2 = \alpha^2}$

Ejercicio 4 Ecuación lineal

$\boxed{\alpha^5 \cdot x + \alpha^{11} = \alpha^2 \iff x = \alpha^4}$

Desarrollo

Ejercicio 1 Cambio de representación y suma

a) $\boxed{\alpha^3 + \alpha^9 = \alpha}$

$$\begin{aligned}
\alpha^3 + \alpha^9 &= \alpha^3 + (\alpha^3 + \alpha) \\
&= (\alpha^3 + \alpha^3) + \alpha \\
&= 0 + \alpha \\
&= \alpha
\end{aligned}$$

b) $\boxed{\alpha^{11} + \alpha^{13} = \alpha^4}$

$$\begin{aligned}
\alpha^{11} + \alpha^{13} &= (\alpha^3 + \alpha^2 + \alpha) + (\alpha^3 + \alpha^2 + 1) \\
&= (\alpha^3 + \alpha^3) + (\alpha^2 + \alpha^2) + \alpha + 1 \\
&= 0 + 0 + \alpha + 1 \\
&= \alpha + 1 = \alpha^4
\end{aligned}$$

c) $\boxed{\alpha^3 + \alpha^9 + \alpha^{11} + \alpha^{13} = 1}$

Verificar el inciso (c) de dos maneras: sumando los cuatro términos de una vez, y sumando los resultados obtenidos en (a) y (b).

| **1.**

$$\begin{aligned}
\alpha^3 + \alpha^9 + \alpha^{11} + \alpha^{13} &= \alpha^3 + (\alpha^3 + \alpha) + (\alpha^3 + \alpha^2 + \alpha) + (\alpha^3 + \alpha^2 + 1) \\
&= (\alpha^3 + \alpha^3 + \alpha^3 + \alpha^3) + (\alpha^2 + \alpha^2) + (\alpha + \alpha) + 1 \\
&= 0 + 0 + 0 + 1 \\
&= 1
\end{aligned}$$

| **2.**

$$\begin{aligned} \alpha^3 + \alpha^9 + \alpha^{11} + \alpha^{13} &= \alpha + \alpha^4 \\ &= \alpha + \alpha + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

d) $\boxed{\alpha^5 + \alpha^5 = 0}$

$$\begin{aligned} \alpha^5 + \alpha^5 &= (\alpha^2 + \alpha) + (\alpha^2 + \alpha) \\ &= (\alpha^2 + \alpha^2) + (\alpha + \alpha) \\ &= 0 + 0 = 0 \end{aligned}$$

Ejercicio 2 Producto y división

a) $\boxed{\alpha^6 \cdot \alpha^{11} = \alpha^2}$

$$\begin{aligned} \alpha^6 \cdot \alpha^{11} &= \alpha^{(6+11)\%15} \\ &= \alpha^{17\%15} \\ &= \alpha^2 \end{aligned}$$

b) $\boxed{\alpha^{12} \cdot \alpha^9 = \alpha^6}$

$$\begin{aligned} \alpha^{12} \cdot \alpha^9 &= \alpha^{(12+9)\%15} \\ &= \alpha^{21\%15} \\ &= \alpha^6 \end{aligned}$$

c) $\boxed{\frac{\alpha^3}{\alpha^{10}} = \alpha^8}$

$$\begin{aligned} \frac{\alpha^3}{\alpha^{10}} &= \alpha^3 \cdot \alpha^{-10} \\ &= \alpha^3 \cdot \alpha^{(15-10)\%15} \\ &= \alpha^3 \cdot \alpha^5 \\ &= \alpha^8 \end{aligned}$$

d) $\boxed{\frac{1}{\alpha^7} = \alpha^8}$

$$\begin{aligned} \frac{1}{\alpha^7} &= \alpha^{-7} \\ &= \alpha^{(15-7)\%15} \\ &= \alpha^8 \end{aligned}$$

e) $\boxed{(\alpha^{14})^3 = \alpha^{12}}$

$$\begin{aligned}(\alpha^{14})^3 &= \alpha^{(14\cdot 3)\%15}\\ &= \alpha^{(42)\%15}\\ &= \alpha^{12}\end{aligned}$$

Ejercicio 3 Expresiones mixtas

a) $\boxed{\dfrac{(\alpha^5+\alpha^9)\cdot\alpha^7}{\alpha^2+\alpha^{12}} = \alpha^6}$

$$\begin{aligned}\frac{(\alpha^5+\alpha^9)\cdot\alpha^7}{\alpha^2+\alpha^{12}} &= \frac{((\alpha^2+\alpha)+(\alpha+\alpha^3))\cdot\alpha^7}{\alpha^2+\alpha^{12}}\\ &= \frac{(\alpha^3+\alpha^2)\cdot\alpha^7}{\alpha^2+\alpha^{12}}\\ &= \frac{(\alpha^3+\alpha^2)\cdot\alpha^7}{\alpha^2+(1+\alpha+\alpha^2+\alpha^3)}\\ &= \frac{(\alpha^3+\alpha^2)\cdot\alpha^7}{1+\alpha+\alpha^3}\\ &= \frac{(\alpha^3+\alpha^2)\cdot\alpha^7}{\alpha^7}\\ &= (\alpha^3+\alpha^2)\cdot 1\\ &= \alpha^6\end{aligned}$$

b) $\boxed{(\alpha^4+1)^2 = \alpha^2}$

Para el inciso (b), resolverlo por dos caminos: primero expresando α^4+1 como potencia de α y elevando al cuadrado; luego, multiplicando los polinomios directamente y reduciendo con la relación $\alpha^4 = 1+\alpha$. Ambos resultados deben coincidir.

| **1.**

$$\begin{aligned}(\alpha^4+1)^2 &= ((\alpha+1)+1)^2\\ &= \alpha^2\end{aligned}$$

| **2.**

$$\begin{aligned}(\alpha^4+1)^2 &= (\alpha^4+1)(\alpha^4+1)\\ &= \alpha^4\cdot\alpha^4+\alpha^4+\alpha^4+1\\ &= \alpha^8+1\\ &= (\alpha^2+1)+1\\ &= \alpha^2\end{aligned}$$

Ejercicio 4 Ecuación lineal

$\boxed{\alpha^5 \cdot x + \alpha^{11} = \alpha^2 \iff x = \alpha^4}$

$$\begin{aligned}
\alpha^5 \cdot x + \alpha^{11} &= \alpha^2 \\
\alpha^5 \cdot x &= \alpha^2 - \alpha^{11} = \alpha^2 + \alpha^{11} \\
x &= \frac{\alpha^2 + \alpha^{11}}{\alpha^5} \\
x &= \frac{\alpha^2 + (\alpha^3 + \alpha^2 + \alpha)}{\alpha^5} \\
x &= \frac{\alpha^3 + \alpha}{\alpha^5} \\
x &= \frac{\alpha^9}{\alpha^5} \\
x &= \alpha^9 \cdot \alpha^{-5} = \alpha^9 \cdot \alpha^{(15-5)\%15} \\
x &= \alpha^9 \cdot \alpha^{10} \\
x &= \alpha^{19\%15} \\
x &= \alpha^4
\end{aligned}$$

Verificar el resultado reemplazándolo en la ecuación original.

$$\begin{aligned}
\alpha^5 \cdot (\alpha^4) + \alpha^{11} &\stackrel{?}{=} \alpha^2 \\
\alpha^9 + \alpha^{11} &\stackrel{?}{=} \alpha^2 \\
(\alpha^3 + \alpha) + (\alpha^3 + \alpha^2 + \alpha) &\stackrel{?}{=} \alpha^2 \\
0 + \alpha^2 + 0 &= \alpha^2
\end{aligned}$$
